

Dive Buddy Alert Band

Buddy communication starts with attention.



Underwater, even a simple message starts with getting the other diver to look. Dive Buddy Alert Band is Rafael's concept for that first moment: a button calls the paired band, a vibration requests attention, and a display shows the response and available distance.

A paired call on the wrist

The proposed interaction begins with a deliberate request for attention. The other diver responds with a button press. During ranging, both screens can show the current separation, with optional vibration cadence for the caller.

This is a communication aid for divers who already use buddy procedures. It does not establish bearing, provide a route, or guarantee reunion.

Concept by Rafael. The CAD Rev D assembly shows the proposed display module. Underwater performance, wearability and manufacturing readiness remain unestablished.

Before a hand signal

The communication problem and the paired-call concept.

Something to share. A buddy looking elsewhere.

A formation in the reef. A change of plan. A routine check-in. The message may be simple, but a hand signal only starts working when the other diver sees it. DBA focuses on the step before that exchange: a direct request for one buddy's attention.

The first request is for attention

The intended use is the moment before a visual exchange: one diver wants the other to look over. Once attention is established, the divers can continue their buddy procedures and visual communication. The band does not attempt to replace the meaning or judgment in that exchange.

A call that belongs to your buddy.

The idea is a paired connection. Before the dive, two bands are linked. During the dive, a deliberate press requests a tactile alert on the other band. The receiving diver presses to acknowledge, so the caller can distinguish a device reply from a person's response.

A visible response

The vibration is intended to interrupt attention briefly; the screen explains why. It shows an incoming call, a buddy's acknowledgement, and fresh separation estimates during a ranging session. If a current reading is unavailable, the number clears rather than lingering as if it were still valid.

Illustrative use scenario. Product performance and user response have not been measured.

User interaction

Proposed surface pairing and underwater button sequence.

1 / Pair on the surface

Hold both buttons for 3 seconds. Each screen shows the paired buddy and local battery status. Address allocation and nearby-pair isolation require firmware definition.

2 / Call

A short press on a ready band calls the paired buddy. B receives three alert pulses and a visible incoming call. A sees Calling, followed by Received by device only after a packet reply.

3 / Acknowledge

B presses once to acknowledge the incoming call. A then sees Buddy acknowledged. A device reply confirms message delivery only. The separate button press communicates the buddy's response.

4 / Range

After the buddy acknowledges, A holds for 1 second to start ranging. Both screens show the latest valid distance and a trend derived from successive readings. A can use the proposed distance cadence alongside the screen.

5 / Stop

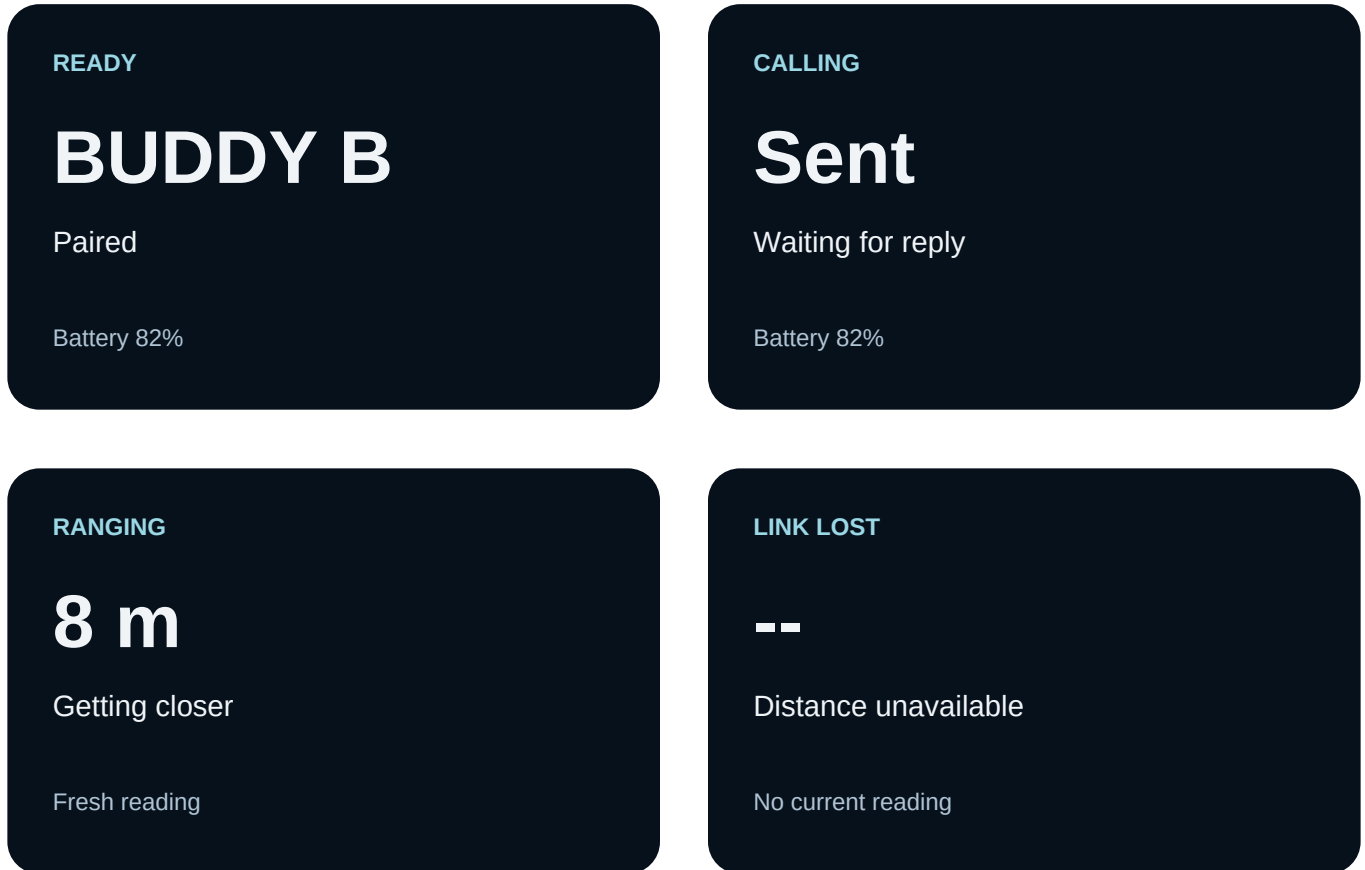
During ranging, a short press on either band requests Stop. The proposed five-minute timeout and one-second burst then stop below 2 m remain. Neither a near reading nor an automatic stop confirms reunion.

6 / Charge

Place the bands on an inductive dock. The display shows local charging status. There is no charging connector on the band. Dock alignment, thermal limits and charge completion require definition.

The display

Proposed information hierarchy. Illustrative values and enlarged screen studies.



A distance requires a fresh measurement

During ranging, the distance takes visual priority. A trend appears only after enough valid readings. A missing or stale reply clears the distance and trend and stops proximity cadence. Reacquisition begins with a fresh reading, without reusing an old trend.

Calling, device receipt and buddy acknowledgement remain separate states. Pair identity and battery stay available. Text accompanies status colors. The display uses the physical button rather than an underwater touchscreen.

These screens specify proposed content and behavior. The 8 m distance and 82% battery are synthetic. Pixel layout, refresh timing, brightness, window size and underwater readability are not established.

Distance feedback

Acoustic range, visual trend and optional caller cadence.

15-20 m / valid reply - One pulse every 4 seconds.

10-15 m - One pulse every 2 seconds.

5-10 m - One pulse every second.

2-5 m - A double pulse every 0.5 seconds.

Below 2 m - One-second burst, then stop. Proximity does not confirm visual contact.

Lost or stale reply - Distance unavailable. Clear the distance and trend and stop proximity pulses.

Distance and orientation

The proposed single transducer estimates separation. A range of 8 m does not establish whether the buddy is left, right, above or below. Relative bearing requires a separate sensing architecture. Exchanging device headings alone would not establish the bearing between divers, and wrist heading need not match swimming direction.

The acoustic exchange

Approximately 65 kHz ultrasonic two-way time-of-flight. Proposed calculation: **$d = (\text{round-trip time} - \text{fixed reply delay}) \times \text{sound speed} / 2$** . At 1,500 m/s with an assumed 2 ms reply delay, 20 m gives approximately 26.7 ms propagation round trip and 28.7 ms total exchange time.

20 m range, approximately ± 0.3 m distance performance and nominal 1 Hz exchange are targets or assumptions. No measured performance is supplied. Timing resolution does not establish accuracy. Multipath, body shadowing and environmental variation remain unresolved.

Made to slip on

Flexible band, rigid electronics module and one reference size.

The band is intended to be flexible silicone: it stretches over the hand, then sits around the wrist or suit cuff. The sealed electronics remain in a rigid, removable module. The circular CAD view shows the band at rest; it does not represent a rigid bracelet.

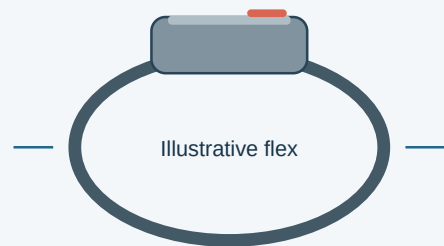
Flexible silicone. Rigid sealed module.

The relaxed CAD shape is one reference size.



At rest

Approx. 195 mm inner circumference



Intended to stretch over the hand

Size coverage and extension remain undefined

Wearability illustration · not a material simulation · relaxed dimensions from CAD Rev D

The fit still needs a size definition

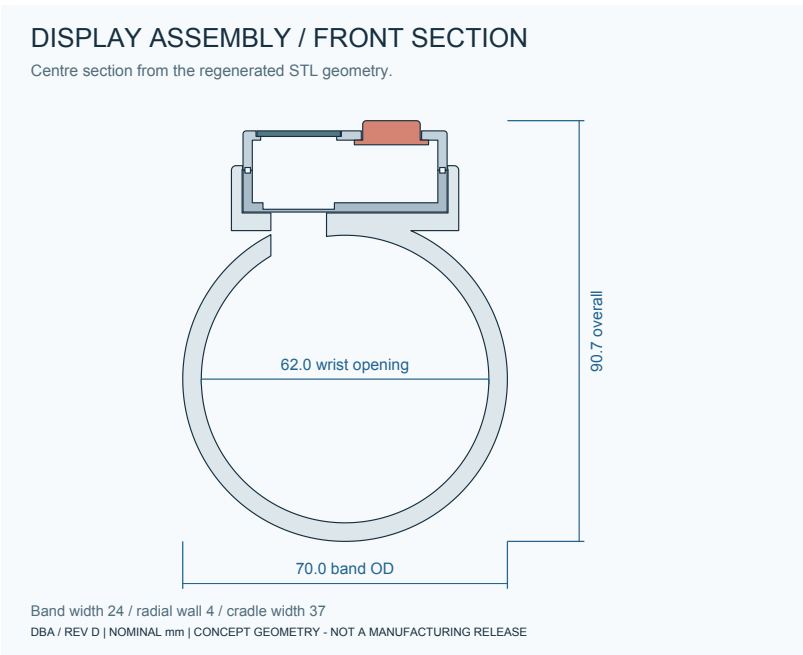
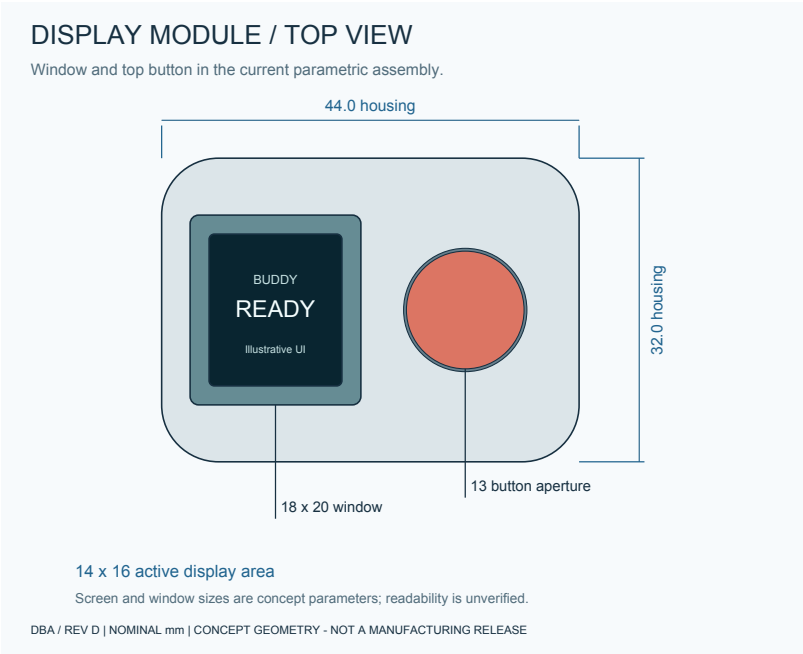
The current model represents one reference size, with a 62 mm relaxed opening and approximately 195 mm inner circumference. A finished size range needs to account for wrist and suit-cuff circumference, hand passage, comfort and module retention. No wrist-size coverage or safe extension has been established.

The nominal band is 24 mm wide with a 4 mm radial wall. The relaxed opening describes the current CAD shape. It does not define a supported wrist or hand size, the stretch needed to put the band on, or the force that would feel comfortable during use.

Silicone grade, hardness, force to stretch, recovery and maximum extension remain undefined. The flex view illustrates intended behavior; it is not a material simulation.

Dimensions & form

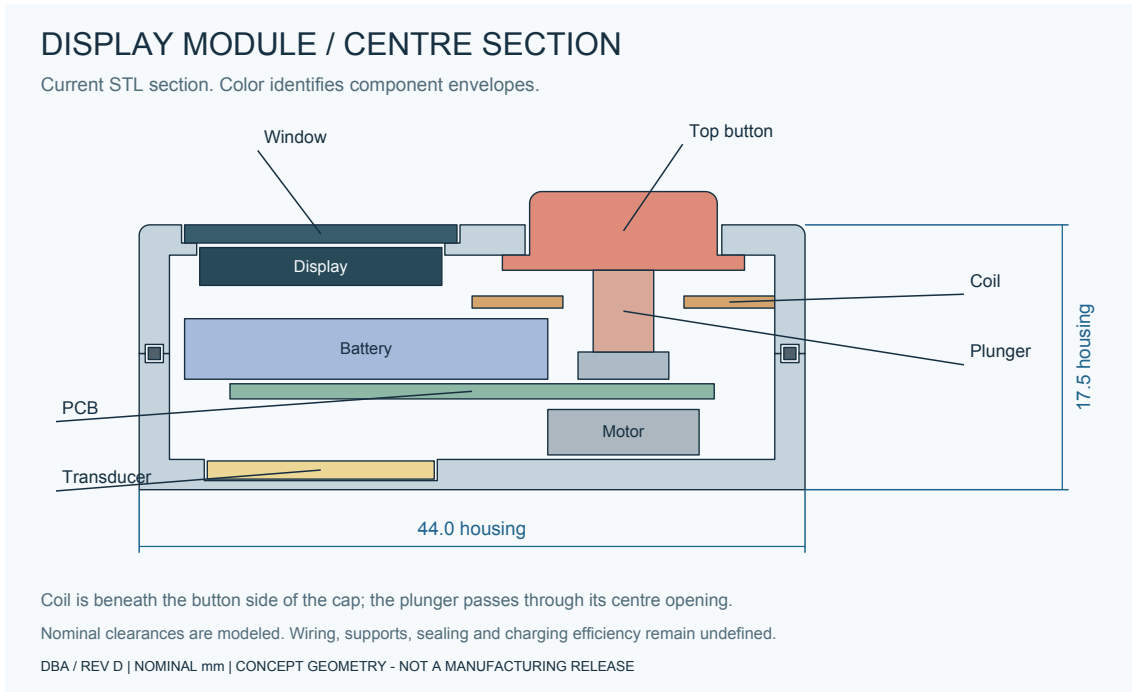
Rev D concept geometry. Nominal dimensions in millimetres, without manufacturing tolerances.



The 44 x 32 x 17.5 mm housing excludes the 2.2 mm button protrusion. Overall assembly height is 90.7 mm. The band retains a 62 mm relaxed opening, 70 mm outside diameter, 24 mm axial width and 4 mm radial wall. These are modeled envelopes, not physical fit or structural verification.

Display & internal layout

Display window, button plunger and receiver-coil arrangement.



Display beside the top button

The display body occupies a nominal 16 x 18 x 2.5 mm envelope beneath an 18 x 20 x 1.2 mm window. The proposed active area is 14 x 16 mm. The display sits on the left of the top cap and the physical button on the right. Optical readability, sealing and electrical interconnects remain unresolved.

Inductive charging through the cap

The receiver coil sits beneath the opaque button side, away from the display window. A nominal 20 mm outside diameter, 8 mm centre opening and 0.8 mm thickness provide a concept envelope around the button plunger. Dock alignment, ferrite, thermal behavior and charging efficiency require engineering definition.

The assembly contains 14 parts, including the display body, window and button plunger. Geometric clearances do not establish complete packaging: seals, adhesives, flex cables, wiring, pressure deflection and assembly access still require development. The sealed module remains removable from the silicone band.

Engineering specification

Register 1-5 of 24. Design decisions, performance targets and pending definition.

Product scope / Fixed decision

One-to-one buddy attention and distance-feedback concept; an aid to buddy procedures, never a safety guarantee.

Pending definition: No promise of reunion, emergency rescue, bearing or navigation.

Discipline: Systems / product

Form and service / Fixed decision

Slip-on silicone band; sealed display module removable from the band; one physical button on top.

Pending definition: Silicone grade, hardness, extension, recovery, wrist/cuff size coverage and hand passage remain undefined. The current circular geometry is relaxed. Retention force, release method and snag loads remain open; removing the module does not mean opening its sealed enclosure.

Discipline: Mechanical / industrial design

Charging interface / Fixed decision

Inductive charging through the sealed cap; no charging port on the band.

Pending definition: The receiver coil is modeled beneath the button side of the cap, beside the display. Dock alignment, magnetic stack, thermal limits, wet/salt-contaminated charging behavior and charge indication remain undefined.

Discipline: Electronics / mechanical

Acoustic method / Fixed decision

Approximately 65 kHz ultrasonic two-way time-of-flight; one wrist transducer.

Pending definition: Transducer construction/resonance, directivity, drive level, link budget and receiver sensitivity need engineering definition. Component names in Rev A are candidates, not selections.

Discipline: Underwater acoustics

Distance and timing / Proposed target

20 m range; approximately ± 0.3 m distance target; nominal 1 Hz exchange with assumed fixed 2 ms reply delay.

Pending definition: No measured performance supplied. Accuracy, resolution, latency and availability need separate definitions. Use $d = (RTT - \text{reply delay}) \times \text{sound speed} / 2$; timing precision alone does not establish accuracy.

Discipline: Acoustics / firmware

Engineering specification

Register 6-10 of 24. Design decisions, performance targets and pending definition.

Operating envelope / Proposed target

Rev A proposes 50 m operating depth, four 60-minute dives per charge and about 28 g mass. These are baseline targets; display-variant mass and runtime are unestimated.

Pending definition: No depth rating or certification established. Define immersion duration, temperature, salinity, pressure convention, storage conditions and intended environments. The Rev A 6 bar note is ambiguous without gauge/absolute pressure.

Discipline: Systems / mechanical

Pairing / Proposed behavior

Hold both buttons 3 s on the surface; show the paired buddy and Ready on both displays. The 16-bit pair code remains a Rev A protocol assumption.

Pending definition: Address allocation, cross-pair isolation and re-pairing protocol remain open. The simulation stages mutual pairing; it does not discover real devices.

Discipline: Firmware

Call and acknowledgement / Proposed behavior

Short press calls the paired buddy. B receives three alert pulses and an incoming-call screen; B short presses to acknowledge. A distinguishes Calling, Device received and Buddy acknowledged.

Pending definition: Transport acknowledgement confirms the device received a packet, not human attention. Retry limits, collision handling and delayed or duplicated messages need protocol definition.

Discipline: Firmware / interaction design

Ranging and stop / Proposed behavior

Caller holds 1 s to request ranging after explicit buddy acknowledgement. Both displays show fresh distance when available; caller cadence is optional. During ranging, a short press on either band stops; nominal 5-minute timeout.

Pending definition: Remote stop requires confirmation. The demonstration separates local stop from unconfirmed delivery when the link is unavailable. Both displays and synchronization remain proposed firmware behavior.

Discipline: Firmware / interaction design

Haptic vocabulary / Proposed behavior

Attention uses three pulses. Optional caller distance cadence retains Rev A: 15-20 m pulse/4 s; 10-15 m pulse/2 s; 5-10 m pulse/s; 2-5 m double pulse/0.5 s; below 2 m a 1 s burst then stop.

Pending definition: Display provides distance without requiring cadence interpretation. Actual intensity, perceptibility through cuffs, hysteresis and waveforms remain unresolved. Near does not mean visual contact or reunion.

Discipline: Interaction design / firmware

Engineering specification

Register 11-15 of 24. Design decisions, performance targets and pending definition.

Lost link / Proposed behavior

Missing or stale replies clear the current distance and trend and suppress proximity pulses. Display reads Distance unavailable; a fresh valid update is required to restore a number.

Pending definition: Demonstration freshness timeout and quality switches are illustrative. Production age limits, validation, outlier handling and reacquisition require acoustic and firmware definition.

Discipline: Systems / firmware

Orientation and echoes / Open question

Body shadowing, wrist orientation, obstructions, other pairs and multipath may affect reception or range.

Pending definition: Define accepted environments, range-quality reporting, outlier handling and stale-reading policy. No directional arrow is provided to the diver.

Discipline: Underwater acoustics

Power budget / Rev A baseline

150 mAh, 3.7 V cell remains a reference envelope. Rev A listening/ranging current and two-hour charging estimates exclude the display and revised module.

Pending definition: Display, illumination and control loads need a revised budget. Rev A stated currents imply about 5.5 mAh for its stated duty cycle before omitted loads, not about 100 mAh. Neither establishes runtime.

Discipline: Electronics

Battery and charge behavior / Proposed behavior

Visible local battery status and a low-battery indication; illustrative demo values do not represent measured charge or a validated reserve.

Pending definition: Low-battery thresholds, reserve behavior, cell protection, cold-water effects, aging and charging completion remain to be defined. Battery percentages alone do not establish remaining dive time.

Discipline: Electronics / firmware

Housing and seal / Rev D concept geometry

44 x 32 x 17.5 mm housing, 2 mm nominal wall and 0.6 mm acoustic membrane. Cap includes a stepped recess for the display window and a top-button aperture.

Pending definition: The model describes solids and clearances. Window attachment, perimeter seal/gland design, pressure deflection, material compatibility, condensation and long-term sealing remain unresolved. No pressure rating or manufacturing tolerances are assigned.

Discipline: Mechanical / acoustics

Engineering specification

Register 16-20 of 24. Design decisions, performance targets and pending definition.

Internal packaging / Rev D concept layout

Fourteen-part assembly includes display body, window and button plunger. The receiver coil sits beside the display; its centre opening surrounds the plunger. Battery and switch occupy separate lateral envelopes.

Pending definition: Envelope geometry does not establish component availability, support features, wire routing, PCB layout, moving-part travel, optical performance or charging efficiency. Bond layers, ferrite and electrical connections require detailed design.

Discipline: Mechanical / electronics

Band and button / Rev D concept geometry

Flexible silicone band shown at rest: 62 mm relaxed inner diameter (approximately 195 mm inner circumference), 24 mm band width and 4 mm radial wall. One reference geometry, with no validated wrist-size coverage. Button aperture 13 mm; protrusion 2.2 mm.

Pending definition: A finished size range must account for wrist and suit-cuff circumference, hand passage, comfort and retention. Silicone grade, hardness, stretch force, recovery and safe extension are unspecified. Button protrusion is not switch travel; glove actuation is unverified.

Discipline: Industrial design / mechanical

Dimensions and drawing authority / Source-derived dimensions

Rev D assembly height is 90.7 mm including the button. Current viewers, annotated drawings and exports use the regenerated source dimensions.

Pending definition: Dimensions describe the concept model; they are not released manufacturing specifications. Rev A/B/C files are retained as legacy references and do not describe the current display assembly.

Discipline: Mechanical

Commercial case / Hypothesis

Proposed first audience: recreational buddy pairs, reached through instructors and dive shops; paired sale with shared dock.

Pending definition: No customer interviews, purchase commitments or defensible novelty evidence supplied. The value of an additional wrist device and the display/haptic combination remains a hypothesis.

Discipline: Product / commercial

Economics / Rev A desk estimate

Original display-free concept: approximately \$43/band, \$86/pair at 5,000 pairs, $\pm 30\%$; \$179-199/pair retail target. Rev D display cost is unestimated.

Pending definition: Baseline only, not a quote or current display-variant cost. Display and revised enclosure/power integration need a new estimate. Tooling, NRE, freight and duty are excluded; margins remain open.

Discipline: Commercial / manufacturing

Engineering specification

Register 21-24 of 24. Design decisions, performance targets and pending definition.

Project status / Presentation Rev E / CAD Rev D

Concept by Rafael. The current presentation combines the product story, proposed display interaction, illustrative silicone flexibility and fourteen-part Rev D parametric assembly.

Pending definition: CAD validity and nominal solid clearances establish model consistency only. Underwater operation, human factors, manufacturing readiness and customer demand are not established.

Discipline: Project lead

Display and readability / Rev D display proposal

Display body 16 x 18 x 2.5 mm; window 18 x 20 x 1.2 mm; active area 14 x 16 mm. Proposed screens show buddy, call state, fresh distance, unavailable data and battery.

Pending definition: These are concept envelopes, not selected display specifications. Whole-metre synthetic demo values do not establish accuracy. Display technology, pixel layout, brightness, viewing angle and underwater legibility require definition.

Discipline: Interaction design / electronics / mechanical

Direction and orientation / Outside current capability

One-transducer two-way timing provides range. No buddy bearing, relative heading or depth difference is measured or displayed.

Pending definition: A directional buddy-finding feature requires separately defined sensing and estimation. Exchanging device headings alone does not determine relative location; wrist heading is not necessarily swimming direction.

Discipline: Systems / underwater acoustics

Distance trend / Proposed behavior

Nearer / farther is an optional textual comparison of successive fresh distance estimates. It describes separation change, not a route or direction to swim.

Pending definition: The simulator uses controlled values. Sampling interval, deadband, stability, quality gating and outlier rejection need definition before a measured trend can be shown.

Discipline: Firmware / interaction design

Designed around the pair

Proposed product, initial audience and commercial assumptions.

A buddy call is useful only when both divers can participate. The proposed product is therefore a pair of bands with one shared inductive charging dock. Recreational buddy pairs are the initial audience; instructors and dive shops are potential places to introduce the concept.

The commercial question is whether an addressed tactile call and readable separation information are valuable enough to earn space on both divers' wrists. The current package defines that proposition; customer demand and the display variant's cost remain unestablished.

Current commercial definition

The display variant has no revised manufacturing cost estimate, retail target or established margin. Its cost must account for the screen and optical window, the revised enclosure, electronics, charging dock, assembly and support. No supplier quotes, customer commitments or market-size claims are supplied.

Historical estimate

Rev A considered the earlier display-free concept at approximately \$43 per band / \$86 per pair, at 5,000 pairs and $\pm 30\%$ uncertainty. Its \$179-199 per-pair retail target is historical. The estimate excluded tooling, non-recurring engineering, freight and duty. These figures do not price the current display assembly.

Engineering scope

The 14-part Rev D parametric concept shows the proposed display module and its nominal component envelopes. Acoustic operation, pressure sealing, electrical integration, wearability and manufacturing readiness still need engineering definition. The display variant has no established runtime or mass.

The original four 60-minute dives per charge and approximately 28 g assembled mass are Rev A reference targets, not demonstrated performance of the current design.

Design files

Current presentation, drawings and parametric source.

DBA_Interactive.html

Primary Rev E product presentation with the dive scenario, paired interaction, flexible-band illustration, Rev D display assembly, dimensions and full specification. Screen values are simulated. Three.js loads from its existing CDN.

DBA_Design_Package_RevE.pdf

Product story, proposed display and button behavior, wearability, technical specification and engineering review notes.

DBA_Product_Overview_RevE.pptx

Twelve-slide product overview with speaker notes. It develops the communication problem, interaction, flexible band and commercial proposition.

DBA_Annotated_Views_RevD.pdf

Three coordinated annotated views of the current display assembly. Nominal concept dimensions include the housing, band, display window and internal section.

cad/, drawings/ and renders/

Regenerated STEP, STL, DXF, drawings and render assets for the 14-part Rev D assembly. Nominal dimensions derive from the parametric source.

dba_cad.py, render.py, drawing_sheet.py

Parametric design and generation scripts. Named parameters define the geometry, which supplies the drawings, renders and embedded meshes.

presentation/

Editable presentation source and requirements register. Regenerated mesh data uses the existing base64 int16 format. The generated design-data.json records dimensions and component envelopes.

Legacy references

Earlier packages remain historical references. Presentation Rev E uses the current story and fit explanation. CAD Rev D governs the unchanged geometry and engineering drawings.

Geometry changes begin in dba_cad.py, followed by regeneration of CAD, renders, drawings and embedded meshes. Keep the model, drawings and presentation aligned.